NAS Guardian

UGREEN AI NAS Hackathon 2026 | Round 1 project proposal | Open Innovation

1. What does the Skill do?

NAS Guardian identifies whether a slow or failing NAS workflow is caused by internet access, a local network link, NAS services, physical hardware/storage, or an externally hosted server, then recommends the safest next check.

2. Who is the target user and problem?

UGREEN NAS owners, home offices, and creators who cannot tell whether poor performance comes from their ISP, Wi-Fi/Ethernet, NAS operating system, failing disks, or an independent cloud-hosted application.

3. What is your proposed solution?

A guided, evidence-first diagnostic assistant producing a five-domain fault matrix, confidence-qualified explanations, safe next tests, and relevant official UGREEN guidance. It explicitly distinguishes an app running on the NAS from one running on a remote origin.

4. What is the core user flow?

The owner describes a symptom; Guardian clarifies whether access is local or remote and where the affected service runs; consults official knowledge; optionally examines authorized files or reports; compares possible causes; then returns a prioritized, non-destructive troubleshooting plan.

5. What makes this different?

Ordinary AI support often conflates a slow internet connection, a slow LAN transfer, a faulty NAS service, a dying disk, and a failed external hosting provider. Guardian separates these layers, labels missing evidence, and avoids fabricated device measurements.

6. Which UGREEN AI NAS capabilities are used and when?

Uliya official knowledge lookup (**ug-knowledge**) supports model-appropriate explanations and troubleshooting. User-authorized read-only filesystem operations (**ug-filesystem**) can inspect selected paths, metadata, and checksums where relevant. Unsupported live sensor/network readings remain user-provided or unavailable.

7. What is the technical approach?

An original Markdown orchestration Skill delegates knowledge questions to **knowledge-agent** and scoped file checks to **file-agent**. A deterministic decision tree tracks five fault domains, evidence provenance, missing observations, approval states, and safe follow-up questions. No credentials, writes, or destructive actions.

8. What is implemented and what remains?

The initial SKILL.md and linked diagnostic, evidence, and validation references are authored and pass the supplied static checker. No NAS runtime testing has occurred. Access to live hardware telemetry is not claimed; finalist testing will establish supported behavior.

9. How will you validate it?

Core case A: slow LAN copies despite normal public internet. Core case B: externally hosted app unavailable while local NAS services function. Edge case: suspected overheating without sensor evidence; Guardian must report uncertainty and avoid unsafe repair steps.

10. What are the limitations and dependencies?

The public Uliya capability catalog lacks direct SMART, temperature, network-speed, and cloud-status measurements. Without approved telemetry or user-provided reports, conclusions are hypotheses, not live diagnostics. Runtime depends on registered Uliya experts and official knowledge availability.

Development disclosure: Original Skill and documentation prepared for this hackathon with OpenAI ChatGPT drafting assistance. No third-party library or binary is bundled; the publicly supplied Uliya environment is a platform dependency. Round 1 is a theoretical pre-test submission, not a claim of working NAS deployment.